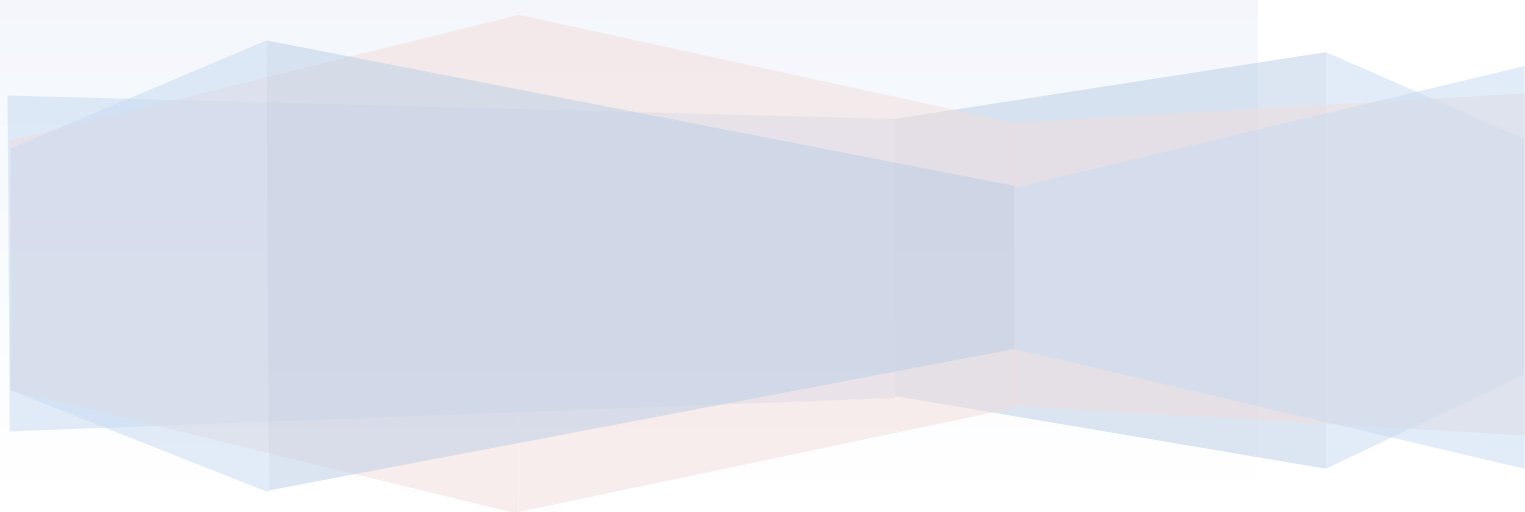


COS30031 Games Programming

Learning Summary Report

Mohamed Shafi Uzman Fassy (102608927)



Self-Assessment Details

The following checklists provide an overview of my self-assessment for this unit.

	Pass (P)	Credit (C)	Distinction (D)	High Distinction (Low HD) (High HD)	
Self-Assessment (please tick)	✓	✓			

Self-assessment Statement

	Included? (tick)
Learning Summary Report	✓
Complete Pass ("core") task work	✓

Minimum Pass Checklist

	Included? (tick)
Additional non-core task work (or equivalent) in a private repository and accessible to staff account.	✓
Spike Extension Report (for spike extensions)	✓
Custom Project plan (for D and/or low HD), and/or High HD Research Plan document (optional)	

Credit Checklist, in addition to Pass Checklist

	Included? (tick)
Custom Project Distinction Plan document, approved	
All associated work (code, data etc.) available to staff (private repository), for non-trivial custom program(s) of own design	
Custom Project "D" level documents , to document the program(s) (structure chart etc) including links to repository areas	

Distinction Checklist, in addition to Credit Checklist

	Included? (tick)
Custom Project "HD" level documents , to document the program(s) (structure chart etc) including links to repository areas	

Low High Distinction Checklist, in addition to Distinction Checklist

	Included? (tick)
High Distinction Plan document, approved	
High Distinction Report document, , which includes links to repository assets	
All associated work (code, data etc.) available to staff (private repository) for your research work	

High High Distinction (Research) Checklist, in addition to D/Low HD Checklist

Declaration

I declare that this portfolio is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature: Mohamed Shafi Uzman

Introduction

This report summarises what I learnt in COS30031 Games Programming. It includes a self-assessment against the criteria described in the unit outline, a justification of the pieces included, details of the coverage of the unit's intended learning outcomes, and a reflection on my learning.

Overview of Pieces Included

This section outlines the pieces that I have included in my portfolio...

- **Task 1: Git Repo Setup**

ULO: Management

In this task I demonstrated my ability to effectively manage a version control system, setting the stage for proper project management and collaborative development practices.

- **Task 2: C++ for Programmers**

ULO: Implementation

In this task I was able to demonstrate and display a solid grasp of C++ programming fundamentals, essential for developing robust game engine components.

- **Task 3: Gridworld**

ULO: Implementation

In this task I was able to address the performance bottlenecks in the Gridworld project I created to highlight my capacity to enhance game efficiency with game loops and separation of update / render code.

- **Task 4: Gridworld Multi-Threaded**

ULO: Implementation & Performance

In this extension tasks I demonstrated my ability to extend Gridworld to showcase my ability to implement multi-threading solutions, significantly optimizing game performance.

- **Task 5: Debugging**

ULO: Implementation, Performance & Maintenance

In this task, I demonstrated my ability to debug complex issues, using different IDE tools.

- **Task 6: Data Structure Basics**

ULO: Design & Implementation

In this task I demonstrated my understanding of various data structures and their appropriate application in game development.

- **Task 7: Performance Measurement**

ULO: Design & Performance

In this task I was able to show by understanding of the topic by establish the importance of collecting and analyzing software performance data using an IDE and changing compiler settings.

- **Task 8: Game State Management**

ULO: Maintenance & Implementation

In this task demonstrated my ability to implement efficient game state management, highlighting my design and implementation abilities.

- **Tasks 9: Game Data Structures**

ULO: Design & Implementation

In this task I demonstrated my ability to research and evaluate appropriate data structures to pick for a particular game design function and behavior.

- **Tasks 10: File Input Output**

ULO: Implementation

In this task I demonstrated my ability to learn how to handle file operations for game data management and persistent storage using JSON library.

- **Tasks 11: Game Graphs from Data**

ULO: Implementation & Management

In this task I demonstrated my understanding of data structures and use of reading from files to create a game world as a graph data structure.

- **Tasks 12: Command Pattern**

ULO: Design, Implementation & Management

In this task I demonstrated my ability to extend existing code to read from file to create game world and process commands.

- **Task 13: Composite and Component Patterns**

ULO: Implementation and Management

In this task I demonstrate my ability to implement composite and component patterns into an existing game world architecture.

- **Task 15: SDL2 Concepts**

ULO: Design & Implementation

In this task I improved my understanding of integrating SDL2 library concepts to add multimedia elements and enhance user interactions with games.

- **Task 16: Sound Board**

ULO: Implementation

In this task I demonstrated my ability in implementing technical audio integration using the SDL library.

- **Task 17: Sprites & Graphics**

ULO: Implementation

In this task I demonstrated my ability to load 2D images from files and display on a window using SDL2.

- **Task 19: Messaging: Announcements & Blackboard**

ULO: Design, Implementation & Maintenance

In this task I demonstrate my ability to create a message-based interaction system where entities and players interact and prompt responses in a game world.

- **Task 22: Collisions**

ULO: Implementation & Design

In this task, I was able to demonstrate the use of physics on 2D games to showcase collision between moving objects.

- **Task 24: Profiling, Performance & Optimisation**

ULO: Performance

In this task I demonstrate my ability to test performance and measure optimization in C++ coding.

- **Task 26: Spike Extension Report**

*ULO: **

In this task I showcase some of the detailed accounts of extensions spikes I have undertaken in the portfolio.

Coverage of the Intended Learning Outcomes

This section outlines how the pieces I have included demonstrate the depth of my understanding in relation to each of the unit's intended learning outcomes.

ILO 1: Design

Discuss game engine components including architectures of components, selection of components for a particular game specification, the role and purpose of specific game engine components, and the relationship of components with underlying technologies.

- **Pass:** In my core spike work, I covered the architectural designs and component selections needed for game specifications. Task 12, where I implemented the command pattern, and Task 13, where I used composite and component patterns, are notable examples.
- **Credit:** I delved deeper into the practical applications of these concepts, especially in Task 9, where I evaluated and chose appropriate data structures for specific game functionalities. My selection was based on a clear understanding of the components' roles and their technological underpinnings.

ILO 2: Implementation

Create games that utilise and demonstrate game engine component functionality, including the implementation of components that encapsulate specific low-level APIs.

- **Pass:** The core spike work provided me with a baseline for working implementations, as seen in Tasks 2 and 3, where I demonstrated fundamental game engine functionality using C++.
- **Credit:** I extended these features in Tasks 4 and 17, where I not only applied multi-threading for one of the tasks but also successfully integrated sprites and graphics using the SDL2 library, showcasing my ability to encapsulate low-level API functionality within game components even though that spike was not considered as an extension.

ILO 3: Performance

Explain and illustrate the role of data structures and patterns in game programming and rationalise the selection of these for the development of a specified game scenario.

- **Pass:** Tasks 7 and 24, where I engaged in performance measurement and optimization, illustrate my ability to explain and showcase the role of data structures and patterns in game programming.
- **Credit:** I provided in-depth analysis and rationalization for my selection of data structures in Task 11, where I created a graph-based game world. This demonstrates my comprehension of the impact of these structures on game performance.

ILO 4: Maintenance

Explain and illustrate the role of data structures and patterns in game programming and rationalise the selection of these for the development of a specified game scenario.

- **Pass:** My explanation and illustration of data structures and patterns were clear in my core spike work, particularly in Task 8, where I showcased a detailed UML graph design for game state management.
- **Credit:** In Task 19, I designed a messaging system that highlighted my understanding of system maintenance. My work showcases the depth of my knowledge and the ability to provide detailed explanations for my design and implementation choices.

Reflection

The most important things I learnt:

From this unit COS30031 I was able to gain a strong understanding of C++, implementing Gameworld State management and Data Structures, and code debugging and performance optimisations that would be useful techniques to apply into my major game development.

The things that helped me most were:

Going to Workshops and doing spike tasks really helped me improve my knowledge and was engaging.

I found the following topics particularly challenging:

I found composite and component patterns a bit challenging in this unit as it expected us to extend our current data structure and implement entities within the game world and create actions and attributes that are interactable. This was complex and time consuming, but I managed to demonstrate and meet the standards that was expected.

I found the following topics particularly interesting:

I found Game Data Structures interesting as it showed us how to read from JSON files that have a data structure for a game world and on how implement them into our gameplay to give users an engaging experience.

I feel I learnt these topics, concepts, and/or tools really well:

I believe I have gained a solid understanding of performance testing, debugging and optimizing in this unit from tasks such as task 5 and 24.

I still need to work on the following areas:

Although I have gained a good understanding of C++ and game data structures, I believe I could explore and learn more to deepen my knowledge in Composite and Component Patterns.

My progress in this unit was ...:

My progress in this unit started off strong but over the weeks I faced some set backs where I struggled on task 5 as I was not too familiar with debugging tools on visual studio since I mostly use visual code and due to this set back I was delayed in working on other tasks which at the end of week 11-12 made me have some tasks left unmarked by the workshop tutor. Which has affected my final grade as I wanted to minimum achieve a Distinction grade. I have learnts to push the challenging tasks for last and focus on completing as many and getting them marked off for my future units.

This unit will help me in the future:

From the knowledge and skills, I have gained from this unit will be invaluable to my future in game development where I could incorporate better practices in coding and performance.

If I did this unit again I would do the following things differently:

- *Time Management: I would manage my time better if I did this unit again, as I found myself completing most tasks after due date and during the late hours of night trying to get them done before the workshop to get them marked off where sometimes I couldn't achieve those goals.*
- *Aimed for a higher grade: I would have gone towards achieving a higher grade by doing all the credit tasks and possibly custom project.*

Other...:

Overall, I am satisfied with my learning experience and progression throughout this unit.

Conclusion

In summary, I believe that I have clearly demonstrate that my portfolio is sufficient to be awarded a low Credit (C) grade.